

PRIYANSHU ROY

AI Systems Engineer & Machine Learning Researcher

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PROFESSIONAL SUMMARY

Third-year B.Tech. Artificial Intelligence & Machine Learning undergraduate and applied researcher specializing in Reinforcement Learning, Large Language Models, and scalable machine learning workflows. Proven track record of architecting production-grade systemsspanning massively parallel physics simulations in NVIDIA Isaac Lab to parameter-efficient model fine-tuning under strict hardware constraints. Driven to solve complex architectural bottlenecks and deploy highly optimized, crash-resilient AI infrastructure.

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, TypeScript, Kotlin, Java, Structured Query Language (SQL), Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, OpenCV, Proximal Policy Optimization (PPO), Generative Adversarial Imitation Learning (GAIL), GANs

LLMs & AI Infrastructure: vLLM, Unsloth, LangGraph, Retrieval-Augmented Generation (RAG), Fine-Tuning (SFT/DPO), NVIDIA Isaac Lab, Omniverse PhysX, Vector Databases (Milvus, pgvector, LanceDB)

Edge AI & Systems: ONNX Runtime, TensorFlow Lite (TFLite), Android Native Development Kit (NDK), Tauri, React, WebAssembly, Linux Architecture

Full-Stack & Developer Tools: Next.js, FastAPI, PostgreSQL, Supabase, Docker, Git, GitHub, Continuous Integration and Deployment (CI/CD)

EDUCATION

Institute of Engineering and Management (IEM) Kolkata, India
Bachelor of Technology in Computer Science (AI & ML) | **CGPA: 8.7/10.0** July 2024 – July 2028 (Expected)

Calcutta Public School Kolkata, India
Class XII (ISC) – Science | **88%** May 2024
Class X (ICSE) – Science | **93%** May 2022

CORE MACHINE LEARNING & RESEARCH

RevExBot | Deep Reinforcement Learning Framework Continuous Control Robotics | March 2026 – Present
Tech Stack: PyTorch (PPO/GAIL), NVIDIA Isaac Lab, Omniverse PhysX, C++

- **Engineered a physically simulated RL framework** in NVIDIA Isaac Lab to train continuous control policies for robotics, contrasting Adversarial Skill Embeddings (ASE) and Mixture of Experts (MoE) paradigms.
- **Simulated 8,192 parallel physical environments** at 50Hz, integrating PyTorch Automatic Mixed Precision (AMP) to execute full Phase 2 GAN training on constrained workstation hardware.
- **Architected a Universal Motion Model (UMM)** featuring a strictly Causal CNN history encoder for temporal momentum, safeguarding against future leakage while protecting manipulator gradients.
- **Enforced training stability** by designing a specialized Teleportation Mask in Generalized Advantage Estimation (GAE) logic to maintain Lipschitz continuity and prevent gradient collapse.

SSENSE | DPDP Alignment Pipeline & SLM Engine Machine Learning & Governance | June 2026 – Present
Tech Stack: Unsloth, TRL, vLLM, Qwen3.5, TypeScript, VS Code API, Bash

- **Architected an end-to-end ML pipeline** to audit Digital Personal Data Protection (DPDP) compliance, utilizing an adversarial GAN Forge (Qwen2-72B) to generate 8,000 highly contextual corporate policy pairs.
- **Trained and aligned a Qwen3.5-9B model** using Unsloth Triton kernels, scaling the maximum sequence length to **24,576 tokens** while avoiding Out-Of-Memory (OOM) panics on Grace Blackwell UMA.
- **Engineered automated VRAM Airlocks** via custom Bash scripting to safely extract 75GB FP8 model weights from active driver contexts, achieving **sub-10 hour** end-to-end Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) alignment.
- **Curated the DPO dataset** by architecting an automated False-Negative dropping mechanism to strictly prevent adversarial data-poisoning.

APPLIED AI & SOFTWARE ENGINEERING

EduBridge | Scalable AI SaaS & Agentic Learning Platform Full-Stack & LLM Orchestration | Feb 2026 – Mar 2026

Tech Stack: Next.js, Supabase, PostgreSQL, Groq, Gemini

- **Designed a generative learning platform** utilizing Gemini to transform static curriculums into personalized, gamified narrative adventures, significantly increasing student engagement.
- **Engineered a low-latency API** powered by Groq LPUs, dynamically routing context to allow the AI tutor to instantly reference the user's exact reading position and mastery level.
- **Optimized database performance** via materialized views, slashing dashboard rendering times to 340ms while maintaining strict multi-tenant data isolation for 50+ concurrent classrooms.

GoPrivate EDR | Zero-Trust Mobile Security Platform Edge AI & Systems Engineering | Jan 2026 – Jun 2026

Tech Stack: Kotlin, C++, JNI, XGBoost, DistilBERT, ONNX, VpnService

- **Architected an on-device machine learning firewall** deploying three synchronized models natively on Android Silicon, eliminating cloud latency and ensuring absolute data sovereignty.
- **Engineered a C++/Java Native Interface (JNI) bridge** to the OS-level VpnService for deep packet analysis, extracting flow heuristics to detect Command and Control (C2) beacons at wire-speed (**<3ms latency, 98.9% F1-Score**).
- **Implemented custom arena allocators** and memory-safe tensor handling, alongside SHAP-based Explainable AI (XAI) to translate complex threat scores into human-readable forensic reports.

HackT | Secure Edge RAG & Threat Intelligence Agent Edge Systems & RAG | Nov 2025 – May 2026

Tech Stack: Tauri (Rust), React, Python, LanceDB, KuzuDB, Quantized LLMs

- **Architected a fully offline, multi-modal RAG agent** utilizing a Tauri frontend to orchestrate local Python microservices, establishing a zero-cloud architecture for highly sensitive threat intelligence.
- **Designed a GPU-aware dynamic VRAM swapping strategy** to seamlessly alternate between vision and language models, operating stably within strict **8GB VRAM constraints**.
- **Built a cryptographic knowledge vault** leveraging hybrid vector-graph retrieval (LanceDB + KuzuDB) with chunked ingestion pipelines, enabling secure semantic search without data exfiltration.

CERTIFICATIONS

- **IBM AI Engineering Professional Certificate** – IBM 2025
- **Goethe-Zertifikat A1: Start Deutsch 1** – Goethe-Institut 2023

ACHIEVEMENTS

- **National Space Hackathon 2026 (IIT Delhi):** Engineered the **Autonomous Constellation Manager (ACM)**, designing autonomous coordination loops and structural multi-agent tracking for satellite constellations.
- **Algorithmic Rigor:** Solved **300+ complex problems** on LeetCode and HackerRank, mastering data structures, dynamic programming, and systems-level memory/runtime optimization.
- **Open Source Contributor:** Actively contribute to public repositories by architecting new features, optimizing execution latency, and resolving critical runtime bugs in edge-AI and security tooling.